MineTrace - Critical Mineral Supply-Chain Risk Assessment

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ANALYST SCOPE NOTES
End-to-end integration verification report.

1. Executive Summary

CLASSIFICATION: INSUFFICIENT_DATA (ANOMALY SCORE: 30.0/100) - PARTIAL
Flag Status: FLAGGED | Sanctions Screening: UNKNOWN | Evaluated Rules: 1 |
Insufficient Rules: 3

DATA INSUFFICIENCY NOTICE
Required baseline trade flows or partner mirror records were missing. Per Phase 3 data sufficiency policies, this transaction cannot be classified as low risk.

Mineral / Commodity:	Lithium (HS 283691)	**Reporter Country:**	USA
Partner Country:	CHL	**Trade Period:**	2023
Score Completeness:	PARTIAL	**Sanctions Screening:**	UNKNOWN

2. Risk Rule Evaluation Details

Rule ID	Rule Name	Weight	Outcome	Threshold	Calculated
R1	Mirror Trade Gap	30 pts	INSUFFICIENT_D..	> 30.0%	N/A
R2	Volume Spike	20 pts	INSUFFICIENT_D..	> 200.0%	N/A
R3	Route Concentration	30 pts	FLAG	> 80.0%	100.0%
R4	Unusual Unit Price	20 pts	INSUFFICIENT_D..	> 60.0%	N/A
R5	Sanctions Exposure Status	0 pts	UNKNOWN	Match against OFA..	N/A

[R1] Mirror Trade Gap (Weight: 30 pts) - Status: INSUFFICIENT_DATA
Formula: abs(export_value - import_value) / max(export_value, import_value) * 100 | Threshold: > 30.0% | Calculated: N/A
Inputs: {'export_value': None, 'import_value': 125000000.0, 'reporter_country': 'United States', 'partner_country': 'Chile', 'period': '2023'}
Explanation: Insufficient data to evaluate mirror trade gap: export value is missing.

[R2] Volume Spike (Weight: 20 pts) - Status: INSUFFICIENT_DATA
Formula: (current_volume - previous_volume) / previous_volume * 100 | Threshold: > 200.0% | Calculated: N/A
Inputs: {'current_volume': 8500000.0, 'previous_volume': None, 'current_period': '2023', 'previous_period': None, 'volume_unit': 'kg'}
Explanation: Insufficient historical data: missing historical previous period baseline.

[R3] Route Concentration (Weight: 30 pts) - Status: FLAG
Formula: partner_import_volume / total_comparable_import_volume * 100 | Threshold: > 80.0% | Calculated: 100.0%
Inputs: {'partner_import_volume': 8500000.0, 'total_comparable_import_volume': 8500000.0, 'partner_country': 'Chile', 'reporter_country': 'United States', 'period': '2023', 'volume_unit': 'kg'}
Explanation: Partner Chile supplies 100.00% (8,500,000.00 kg) of total comparable imports (8,500,000.00 kg) for United States in 2023, exceeding the 80.0% concentration threshold.

[R4] Unusual Unit Price (Weight: 20 pts) - Status: INSUFFICIENT_DATA
Formula: abs(unit_price - comparable_median_price) / comparable_median_price * 100 | Threshold: > 60.0% | Calculated: N/A
Inputs: {'trade_value': 125000000.0, 'net_weight_kg': 8500000.0, 'calculated_unit_price': 14.7059, 'sample_size': 1, 'min_required_sample_size': 3, 'commodity_code': '283691'}
Explanation: Insufficient comparable records: found 1 valid transactions, but at least 3 are required to establish a robust median benchmark.

[R5] Sanctions Exposure Status (Weight: 0 pts) - Status: UNKNOWN

Formula: Exact or normalized string matching against official OFAC SDN names and aliases | Threshold: Match against OFAC SDN List (Separate from numerical anomaly score) | Calculated: N/A
Inputs: {'query_entity_name': None, 'available_ofac_records_count': 3}
Explanation: UNKNOWN: Source trade record provides country-level aggregate trade data and contains no specific company or entity identifier suitable for sanctions screening. Sanctions screening cannot be performed without entity identification.

INSUFFICIENT RULES EXPLANATION
The following rule(s) could not be evaluated due to missing source data or baselines: R1, R2, R4. Under MineTrace sufficiency policy, un-evaluated rules are never treated as passing.

3. Classification & Sufficiency Guardrails

Saved Anomaly Score:	30.0 / 100.0	**Assigned Tier:**	INSUFFICIENT_DATA
Flag Status:	FLAGGED	**Policy Version:**	1.0

Thresholds Applied: GREEN (0-19), AMBER (20-49), RED (50-100), INSUFFICIENT_DATA (PARTIAL/UNAVAILABLE)
Classification Rationale: INSUFFICIENT_DATA: Score completeness is PARTIAL because one or more scoring rules (R1, R2, R4) could not be evaluated. By data sufficiency policy, incomplete assessments cannot be assigned a conclusive color tier. Numerical score (30.0) is preserved for auditability.

4. Sanctions Screening (OFAC SDN)

SANCTIONS STATUS: UNKNOWN
Registry Reference: OFAC Specially Designated Nationals (SDN) List Limitations: SANCTIONS SCREENING LIMITATIONS: Screening is performed against public US Treasury OFAC Specially Designated Nationals (SDN) registries. Country-level or commodity-level screening does not substitute for corporate beneficial ownership checks (e.g., 50 Percent Rule) or end-user certification.

5. Source Evidence & Audit Trail

Source Dataset:	UN Comtrade (Normalized)	**Source Record ID:**	COMTRADE-REC-001
Retrieval Timestamp:	2026-10-27T16:55:28.413327	**Evidence Snapshot ID:**	00dd0302-230f-42bf-887b-bafc4a3cb1b5
Validation Status:	VALID	**Source Endpoint / URL:**	https://comtradeapi.un.org/public/v1/pre

Event ID	Event Type	Status	Timestamp
07c4ef5f-f3c..	RAG_QUERY_EXECUTED	SUCCESS	2026-10-27T16:55:29
19d44041-939..	EVIDENCE_VALIDATION_PASSED	SUCCESS	2026-10-27T16:55:28
9807e958-392..	CLASSIFICATION_CREATED	SUCCESS	2026-10-27T16:55:28
9d2ee1bd-1e9..	RISK_ANALYSIS_CREATED	SUCCESS	2026-10-27T16:55:28
9c270a7a-87d..	EVIDENCE_SNAPSHOT_GENERATED	SUCCESS	2026-10-27T16:55:28

6. Validated AI Explanation

Status: UNAVAILABLE. No pre-validated AI explanation was attached to this transaction record.

7. Methodology, Limitations & Legal Notice

NOTICE & DISCLAIMER: This document is an automated evidence report generated by the MineTrace critical mineral supply-chain risk intelligence system. Numerical anomaly scores, rule evaluations, and risk classifications (GREEN, AMBER, RED, INSUFFICIENT_DATA) reflect deterministic statistical and relational anomalies compared against historical baselines and official trade registries. They DO NOT constitute a legal finding of fraud, illicit trade, customs violation, or sanctions evasion. A GREEN classification does not certify a supply chain as compliant, ethical, or risk-free. Missing or incomplete data must never be interpreted as low risk.

METHODOLOGY & LIMITATIONS:

1. Anomaly Score: A deterministic weighted score (0-100) aggregating multi-rule trade indicators.
2. Scope of Detection: Risk rules evaluate reported trade flows (Comtrade) and public SDN lists (OFAC). They do not inspect physical shipping containers or corporate beneficial ownership trees.
3. Data Sufficiency: Where required historical baselines or partner trade reports are unavailable, MineTrace strictly enforces sufficiency guardrails and assigns INSUFFICIENT_DATA rather than assuming compliance.
4. Sanctions Screening: Evaluates exact and normalized name matches against US Treasury OFAC records. A status of NO_IDENTIFIED_MATCH does not verify compliance with non-US regimes or unlisted intermediaries.
5. Prototype Status: Thresholds (0-19 GREEN, 20-49 AMBER, 50-100 RED) are heuristic operational baselines designed for screening and prioritization, not statistically validated probabilities of wrongdoing.